MOMINA PARSA

Software Engineering Student | AI Engineering

Islamabad, Pakistan | +92 343 5909901 | momina109malik@gmail.com
GitHub: github.com/momina-eng | LinkedIn: linkedin.com/in/momina-parsa

PROFILE

Software Engineering student with strong foundations in data structures, object-oriented programming, software design, and application development. Experience spanning web applications, database systems, semantic search, and AI-powered software solutions. Currently contributing as an AI Engineering Intern at a government organization on applied AI and software engineering workflows, with a focus on continuous learning and practical problem-solving.

TECHNICAL SKILLS

Programming: Java, Python, C/C++, JavaScript, SQL
Development: HTML5, CSS3, JavaScript, Node.js, Express.js, REST APIs
AI & Search: AI-Powered Software Development, Google Gemini, Semantic Search, Vector Embeddings, Sentence Transformers
Frameworks: React, FastAPI, LangGraph, MONAI
Databases: MongoDB, Oracle SQL, ChromaDB, FAISS
Tools: Git, GitHub, VS Code, Linux, Oracle APEX
Core: Data Structures & Algorithms, Object-Oriented Programming, Software Design, Database Systems

PROFESSIONAL EXPERIENCE

AI Engineering Intern July 2026 – Present
Government Organization

- Contribute to AI-powered software solutions and support applied AI and software development workflows.
- Collaborate with engineers on real-world software problems across design, implementation, and iteration, integrating AI into production-oriented systems.

PROJECT EXPERIENCE

AI Vulnerability Analysis & Automated Patching

Python · FastAPI · Google Gemini · React

- Engineered an AI-assisted source-code analysis system that identifies vulnerabilities and proposes automated fixes.
- Combined vulnerability classification, CWE mapping, remediation retrieval, and Google Gemini–based patch generation with deterministic validation and iterative refinement.

Semantic News Retrieval Platform

Python · FastAPI · Sentence Transformers · ChromaDB · FAISS · Gradio

- Developed a semantic retrieval platform for discovering contextually relevant Pakistani news articles.
- Applied multilingual Sentence Transformers embeddings with ChromaDB and FAISS for similarity search, exposed through a Gradio interface.

University Feedback Management System

HTML · CSS · JavaScript · Node.js · Express.js · MongoDB

- Created a web platform for submitting, tracking, and managing university complaints and feedback.
- Built Node.js/Express.js REST APIs with MongoDB persistence and structured administrative review workflows.

Healthcare Appointment & Management System

Oracle APEX · Oracle SQL

- Built a database-driven healthcare appointment management application using Oracle APEX.
- Designed relational data structures and implemented CRUD workflows with Oracle SQL, including interactive dashboards for data management.

Rental Management Application

Java · Java Swing · Object-Oriented Programming

- Developed a Java-based rental management application using object-oriented design principles.
- Implemented workflows for tenant records, property tracking, and rental bookings through a Java Swing interface.

EDUCATION

Bachelor of Science in Software Engineering Expected Graduation: January 2028
COMSATS University Islamabad

LEADERSHIP

Academic Lead — CSES Software Engineering Society, COMSATS University Islamabad
Co-Lead — GDG Google Developer Group On Campus
Subcircle Lead Software Engineering Research Sub-Circle, Research Circle

RELEVANT COURSEWORK

Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, Web Technologies, Operating Systems, Software Requirements Engineering, Design & Analysis of Algorithms, Artificial Intelligence